CSE 301

IRK Class Notes

Adapted from Rifah Tasnia Islam

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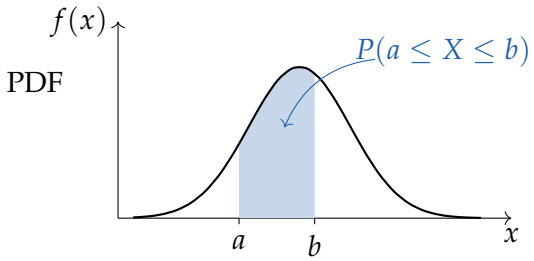
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Lecture 1 PDF, CDF & Parametric Models

15/08/26

PDF vs. CDF



- Area under the PDF between a and b gives $P(a \le X \le b)$.
- $f(5)$ is *not* a probability: $f(5) \neq P(X = 5)$ (for a continuous r.v. $P(X = 5) = 0$).
- **Think — dartboard fallacy:** probability of hitting one exact point is $\frac{1}{\infty}$ (i.e. $\frac{\text{finite}}{\infty} \to 0$), yet some point always gets hit.

Definition (CDF). CDF $\to F(x)$ — integration এর value precomputed; this is what we actually mean by "*distribution*". **We focus on the CDF.**

Parametric vs. Non-parametric Model

- **Parametric:** finite সংখ্যক ধ্রুবক (constants) দিয়ে model describe করা যায়.
- **Non-parametric:** infinite সংখ্যক ধ্রুবক লাগে।

Those constants are called **"Parameters"**.

- Bernoulli $\to$ 1 parameter (p)
- Binomial $\to$ 2 parameters (n, p)
- Normal $\to$ 2 parameters (μ, σ)

"Sampling" করে real distribution এর parameters guess করি $\to$ ML এর *training data*.

Sample vs. Population

Fair coin (H/T) can be modeled by a Bernoulli distribution. Now someone has to guess whether the coin is fair.

Samples $\to [H, T, T, H, H, H]$
Population $\to \text{Bern}(\frac{1}{2})$
Both are arrays. Population: infinite size এর array (of H & T) — the true distribution, $\#H = \#T$.

Sample দেখে মনে হবে 67% (H), 33% (T). But actually it's 50–50. Here 6 is the *sample size*.

Target. Sample দেখে Population বুঝতে পারা $\implies$ এমন একটা func./algo. দেয়া, যেটা sample size যত বেশি তত ভালো কাজ করে।

Lecture 2 Estimators, Bias, Standard Error, Convergence 16/08/26

Book: Wasserman, *All of Statistics*.

- θ : actual parameter of the underlying distribution → **fixed**.
- $\hat{\theta}$: guessed parameter → data dependent, so a **random variable**.
- $\hat{\theta}_n$: কত sample size এর জন্য guess করব (n = sample size).

Note. θ-কে constant হিসেবে treat করতে পারব, $\hat{\theta}$-কে না।

$$P(X \in A) = \int_A f(x)\,dx, \qquad \theta \in \Theta \text{ (parameter space; e.g. Bernoulli} \to [0,1]).$$

$P_\theta(X \in A)$: subscript means the probability under parameter θ. E.g. variance বের করব, H উঠার chance $p = 20\% \to$ লিখি $V_{0.2}$.

Estimator

$$X_1, X_2, \ldots, X_n \overset{\text{iid}}{\sim} F, \qquad \hat{\theta}_n = g(X_1, X_2, \ldots, X_n) \quad \text{(guess)}$$

- $g \to$ **Estimator** (anything from a basic avg. to a complex neural network)
- $\hat{\theta}_n \to$ **Estimate**

Bias

Definition (Bias).

$$\text{bias}(\hat{\theta}_n) = \mathbb{E}_\theta(\hat{\theta}_n) - \theta$$

actual থেকে guess গুলো কতটুকু দূরে। $= 0$ হলে guesses *average out*.

Example. 6 বার coin flip → $p = 67\%$; next দিন, 6 বার coin flip → $p = 55\%$; ... অনেকবার repeat করলে actual-এর কাছে যাওয়া উচিত।

Biased: g func-টা more biased towards fair coin. তাহলে unfair coin দিয়ে test করলেও যতই avg নেই, avg out হয় না! The algo/estimator is *biased*.

Sampling Distribution & Standard Error

$\hat{\theta}_n$ একটা r.v. এটার distribution → **sampling distribution** (estimate-গুলোর distribution). এর S.D.-কে বলব **standard error**:

$$\text{se}(\hat{\theta}_n) = \sqrt{\mathbb{V}(\hat{\theta}_n)}$$

— guess গুলো একে অপর থেকে কতটুকু দূরে।

Note. Actual param. θ না জেনেও bias বের করব, how?

Dartboard analogy: dartboard এর bull's eye → θ; bias: dartboard এর যেকোন দিকে throws বেশি।

se ↓, bias ↑	possible	We expect both ↓.
se ↑, bias ↓	possible	

Convergence in Probability

Definition. $X_n \xrightarrow{P} X$ if for every $\epsilon > 0$,

$$P\big(|X_n - X| > \epsilon\big) \to 0 \quad \text{as } n \to \infty.$$

Here $X_n \leftrightarrow \hat{\theta}_n$, $X \leftrightarrow \theta$, n = sample size (not strictly). যেকোন positive no. এর জন্য guess & actual এর distance ঐ no.-এর চেয়ে বড় হওয়ার probability $\to 0$ as $n \to \infty$.

Convergence in Quadratic Mean

Definition. $X_n \xrightarrow{qm} X$ if $\mathbb{E}(X_n - X)^2 \to 0$ (avg squared diff).

$X_n \xrightarrow{qm} X$ implies that $X_n \xrightarrow{P} X$.

Consistency

Definition. $\hat{\theta}_n$ is **consistent** if $\hat{\theta}_n \xrightarrow{P} \theta$.

bias কম হলে consistent হয় কিনা? vice-versa? $\Rightarrow$ **Prefer consistency over unbiasedness.**

Example. $X_1, \ldots, X_n \sim \text{Bern}(p)$, $\hat{p}_n = n^{-1}\sum_i X_i$. Biased?

$$\mathbb{E}\left[\frac{\sum X_i}{n}\right] = \frac{1}{n}\sum_i \mathbb{E}[X_i] = \frac{np}{n} = p$$

$\therefore$ $\hat{p}_n$ is unbiased.

Questions

1. Given an **unbiased** estimator, how can you use this to find the actual parameter?
   — অনেকগুলো experiment এর result avg নিবো। Sample size not necessarily big.
2. Given a **consistent** estimator...?
   — experiment 1টাই, অনেক বড় sample, $n \to \infty$.

Lecture 3 MSE, Consistency & Confidence Intervals 17/08/26

Mean Squared Error

$$\text{MSE} = \mathbb{E}_\theta(\hat{\theta}_n - \theta)^2 \qquad \text{and} \qquad \boxed{\text{MSE} = \text{bias}^2(\hat{\theta}_n) + \mathbb{V}_\theta(\hat{\theta}_n)}$$

where $\mathbb{V}_\theta(\hat{\theta}_n) = \text{se}^2$. (Proof $\to$ slide.) MSE $\downarrow$ মানে bias $\downarrow$, se $\downarrow$.

Showing that an Estimator is Consistent

$\hat{\theta}_n \xrightarrow{P} \theta$ সরাসরি দেখানো ঝামেলা। So we use a **sufficient condition**:

$$\text{bias, se} \to 0 \ (n \to \infty) \implies \text{MSE} \to 0 \implies \hat{\theta}_n \xrightarrow{qm} \theta \implies \hat{\theta}_n \xrightarrow{P} \theta.$$

(MSE is exactly $\mathbb{E}(\hat{\theta}_n - \theta)^2$, the quadratic-mean quantity.)

Caution. But bias $\to 0$, se $\to 0$ না হলে (MSE $\to 0$ না হলে) it is *not necessarily inconsistent* — consistent হতে পারে।
If "bias $\to 0$, se $\to 0$" a *necessary* condition হতো, তাও বলতাম consistent means unbiased $\to$ **Wrong!** (bias $\to 0$ is not the same as bias $= 0$.) Recall: $P \to Q$ does not give $\neg P \to \neg Q$.

Recipe: $\hat{\theta}_n$ দেয়া, consistent কিনা check. bias বের করব, se বের করব $\to$ limit নিয়ে দেখব, $n \to \infty$ তে 0 হয় কিনা।

Example (Unbiased but inconsistent). $X_1, \ldots, X_n$, estimator $\hat{\mu} = X_1$.

$$\text{Bias} = \mathbb{E}(\hat{\mu}) - \mu = \mathbb{E}(X_1) - \mu = \mu - \mu = 0.$$

Intuition: বারবার experiment করলে avg out হয়। কিন্তু $n \to \infty$ হলেও just 1টা experiment-এ, accuracy বাড়ে না। So, **inconsistent**.

Confidence Interval

Search area (interval of points) for the parameter: (a, b) — a, b are *data dependent values*.

- α : tolerance for error
- $1 - \alpha$: at least যতটুকু confident — called the **coverage** / **confidence level**.

A $(1 - \alpha)$ confidence interval C_n satisfies $P_\theta(\theta \in C_n) \geq 1 - \alpha$.

For vector $\theta \to$ multiple params $(\mu, \sigma, \ldots)$: $(1, 2)$, $(1.1, 2.5), \ldots$ — a set of points / geometric area.

Interpretation — C_n is random, θ is fixed.

- "95% probability that θ will fall in C_n" ✗ (this treats θ as random and C_n as fixed — both wrong)
- "Interval গুলোর (C_n, random) মধ্যে 95% interval-এ θ (fixed) থাকে" ✓

Best interpretation: "95% of all 95% CIs hold the true parameter."
Analogy: সবাই 60% $P(H)$ coin toss করলে total head উঠার chance 60% — 60% of the 60% coins came up heads.

Lecture 4 Confidence Sets (example), Normal-based CI, MLE 22/08/26

Example (A confidence set). X_i $+1$ or -1 হতে পারে, $P(X_i = \pm 1) = \frac{1}{2}$ — we can't see it. "Data": $Y_i = \theta + X_i$ — Y_1, Y_2 we can see (observable).

Confidence set: single element set হবে. If $Y_1, Y_2 = 15, 17$:

$$\frac{15 + 17}{2} = \{16\} = \theta \ \to \ 75\% \text{ C.S.}$$

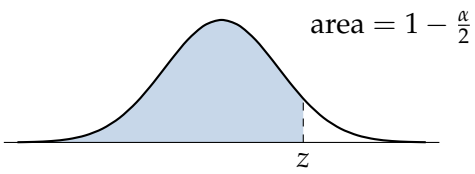
(the set is built as a *function of the data*). X_1, X_2 এর 4টা condition এর মধ্যে 3টা case-এ correct:

X_1	X_2
$+1$	$+1$
$+1$	-1
-1	$+1$
-1	-1

Normal-based Confidence Interval

$$\Phi(u) = P(Z < u), \qquad z_{\alpha/2} = \Phi^{-1}\left(1 - \tfrac{\alpha}{2}\right)$$

Z এর কোন value পর্যন্ত probability $\left(1 - \frac{\alpha}{2}\right)$ হবে।



$$C_n = \left(\hat{\theta}_n - z_{\alpha/2}\,\widehat{\text{se}},\ \hat{\theta}_n + z_{\alpha/2}\,\widehat{\text{se}}\right), \qquad P(\theta \in C_n) \approx 1 - \alpha$$

Normal distribution হওয়া জরুরি না — approx./imaginary Normal (the *sampling* distribution). Sampling distribution এর mean যদি θ হয়ে থাকে,

$$Z_n = \frac{\hat{\theta}_n - \theta}{\widehat{\text{se}}}.$$

Unbiased dist. given, CI বানাব normal থেকে।

Note (Typical problem). Situation + কিছু data → guess given (formula) → সেই estimator biased? MSE, se, consistent? CI, 95% CI (table থেকে data নিয়ে, 5% error).

Maximum Likelihood Estimators (MLE)

Data দেখে parameters জানতে চাই। Parameter controls the data [not vice-versa]. We have data, so we want to guess the parameters:

- $P(\theta \mid \text{data})$ — কোন param এর prob. কত: **Posterior Probability**.
- $P(\theta)$ — I can control it. With prior knowledge, I can make θ_2 more likely: **Prior Probability**. It affects $P(\text{data} \mid \theta)$.

(AI → hypothesis guess; now → param. guess — same idea.)

$$\therefore\ P(\theta \mid \text{data}) \ \propto\ P(\text{data} \mid \theta) \qquad [\text{বাকি জিনিস constant}]$$

If we want to maximize the left side, we can just maximize the right side → **MLE**: কোন θ এর জন্য max পাব।

Maximize করতে → differentiation.

Note. MLE → TF-এ surely 1টা set থাকবে।

Lecture 5 Likelihood Function & MLE Examples

23/08/26

Definition (Likelihood function).

$$\mathcal{L}_n(\theta) = \prod_{i=1}^{n} f(X_i;\theta) = P(\text{data} \mid \theta)$$

(data and param are separated by ";").
Log-likelihood: $\ell_n(\theta) = \log \mathcal{L}_n(\theta) \to$ differentiation easier.

Definition (MLE). The MLE $\hat{\theta}_n$ is the param. for which the likelihood function is maximized — the parameter that makes the data সবচেয়ে believable.

Bernoulli

$$f(x) = \begin{cases} p, & x = 1 \\ 1-p, & x = 0 \end{cases} \implies f(x;p) = p^x(1-p)^{1-x}, \quad x = 0,1$$

$$\mathcal{L}_n(p) = \prod_{i=1}^{n} p^{X_i}(1-p)^{1-X_i} = p^S(1-p)^{n-S}, \qquad S = \sum_{i=1}^{n} X_i$$

$$\ell_n(p) = S\log p + (n-S)\log(1-p) \quad \text{(for which } p \text{ is } \ell_n(p) \text{ max?)}$$

$$\ell_n'(p) = \frac{S}{p} - \frac{n-S}{1-p} = 0 \implies \boxed{\hat{p}_n = \frac{S}{n}} \text{ (avg)}$$

MLE $\to$ intuitive result.

Normal

$X_1, \ldots, X_n \sim N(\mu, \sigma^2)$ (μ: population mean, σ^2: population variance).

$$\bar{X} = n^{-1}\sum_i X_i \text{ (sample mean)}, \qquad S^2 = n^{-1}\sum_i (X_i - \bar{X})^2 \text{ (sample variance)}$$

$$\mathcal{L}_n(\mu,\sigma) = \prod_{i=1}^{n} \frac{1}{\sigma} e^{-\frac{1}{2\sigma^2}(X_i-\mu)^2} = \sigma^{-n} e^{-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(X_i-\mu)^2}.$$

Write $(X_i - \mu)^2 = (X_i - \bar{X} + \bar{X} - \mu)^2$; summing, the cross term $2ab$ vanishes, so $\sum_i (X_i-\mu)^2 = nS^2 + n(\bar{X}-\mu)^2$:

$$\mathcal{L}_n(\mu,\sigma) = \sigma^{-n} e^{-\frac{nS^2}{2\sigma^2}} e^{-\frac{n(\bar{X}-\mu)^2}{2\sigma^2}} \qquad \text{(do detailed steps yourself)}$$

$$\frac{\partial \ell(\mu,\sigma)}{\partial \mu} = 0, \quad \frac{\partial \ell(\mu,\sigma)}{\partial \sigma} = 0 \implies \boxed{\hat{\mu} = \bar{X}, \quad \hat{\sigma} = S}$$

(S = data গুলোর std. deviation).

Uniform

$X_1, X_2, \ldots, X_n \sim \text{Uniform}(0, \theta)$.

Informal. Data $1, 10, 9, 3$: $\theta \geq 10$; $\theta \to 20$? $999 \to$ unlikely. So $\theta \to 10 = \max(X_1, \ldots, X_n)$, i.e. $\hat{\theta} = X_{(n)}$ — X array-কে non-decreasing order এ সাজানোর পর n-th index = max.

Formal. $U(a, b)$ has PDF $= \frac{1}{b-a}$ [area $= 1$]. Unif$(0, \theta)$ এর জন্য $f(x; \theta) = \frac{1}{\theta}$.

$$\mathcal{L}_n(\theta) = \begin{cases} \left(\frac{1}{\theta}\right)^n, & \theta \geq X_{(n)} \\ 0, & \theta < X_{(n)} \quad \text{(impossible case)} \end{cases}$$

$\theta \uparrow \Rightarrow \mathcal{L}_n(\theta) \downarrow$. $\therefore$ $\mathcal{L}_n(\theta)$ max পাব first value $X_{(n)}$ এর জন্য: $\boxed{\hat{\theta} = X_{(n)}}$.

Exercise (DIY)**.** Unif(a, b): $\mathcal{L}_n(a, b) = \left(\frac{1}{b-a}\right)^n$. Find the MLE.

Lecture 6 Bias of the MLE & Bessel's Correction 24/08/26

MLE: good, but **not always unbiased**.

Mean. $\hat{\mu} = \bar{X}$ is unbiased:

$$\mathbb{E}[\hat{\mu}] = \mathbb{E}[\bar{X}] = \mathbb{E}\left[\frac{\sum X}{n}\right] = \frac{\sum \mathbb{E}[X]}{n} = \frac{n\mu}{n} = \mu \quad \text{(population mean)}.$$

Variance. $\hat{\sigma}^2 = S^2 = \frac{1}{n}\sum_{i=1}^n (X_i - \bar{X})^2$ is **biased**:

$$\mathbb{E}[S^2] = \frac{n-1}{n}\sigma^2 \neq \sigma^2.$$

Derivation (slide-এ steps):

$$\begin{aligned} \mathbb{E}[S^2] &= \frac{1}{n}\sum_{i=1}^n \mathbb{E}\left[\left((X_i - \mu) - (\bar{X} - \mu)\right)^2\right] \\ &= \frac{1}{n}\sum_{i=1}^n \mathbb{E}[(X_i - \mu)^2] - \frac{2}{n}\mathbb{E}\left[(\bar{X} - \mu) \cdot n(\bar{X} - \mu)\right] + \mathbb{E}[(\bar{X} - \mu)^2] \end{aligned}$$

using $\sum_i (X_i - \mu) = \sum X_i - \sum \mu = n(\bar{X} - \mu)$. The $2ab$ and b^2 terms মিলে merged into one b^2:

$$\mathbb{E}[S^2] = \sigma^2 \underbrace{-\mathbb{E}[(\bar{X} - \mu)^2]}_{\text{bias term}}$$

The 2nd term is (কোন data – তার avg)^2 এর avg = ঐ data এর variance:

$$\mathbb{E}[(\bar{X} - \mu)^2] = \mathbb{V}(\bar{X}) = \mathbb{V}\left(\frac{1}{n}\sum_1^n X_i\right) = \frac{1}{n^2}\sum_1^n \mathbb{V}(X_i) = \frac{1}{n^2}(n\sigma^2) = \frac{\sigma^2}{n}.$$

$$\therefore \mathbb{E}[S^2] = \sigma^2 - \frac{\sigma^2}{n} = \frac{n-1}{n}\sigma^2, \qquad \text{bias}(S^2) = -\frac{\sigma^2}{n}.$$

The estimator is biased, and thus **MLE can be biased too**.

How to resolve? — Bessel's Correction. Divide by $(n-1)$ instead of n:

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2 \;\to\; \text{unbiased.}$$

As $n \to \infty$, the $-\frac{\sigma^2}{n}$ term becomes negligible. So, sometimes the original estimator is used.

Consistency of the MLE — Proof Sketch

$X_1, X_2, \ldots, X_n \sim f(x;\theta_0)$, where θ_0 is the **true parameter**.

$$\begin{aligned}
\text{MLE, } \hat{\theta}_n &= \arg\max_{\theta} \prod_{i=1}^{n} f(X_i;\theta) && \text{(likelihood)}\\
&= \arg\max_{\theta} \sum_{i=1}^{n} \log f(X_i;\theta) && \text{(log-likelihood)}\\
&= \arg\max_{\theta} \frac{1}{n}\sum_{i=1}^{n} \underbrace{\log f(X_i;\theta)}_{\text{random}}
\end{aligned}$$

অনেকগুলো r.v.-কে avg করলে ঐ avg-টা true avg এ যায়, as $n \to \infty$ (LLN). Define

$$L_n(\theta) = \frac{1}{n}\sum_{i=1}^{n} \log f(X_i;\theta) \quad (n\text{-sized sample এর avg}),$$

which goes towards the true avg of $\log f(X_i;\theta)$:

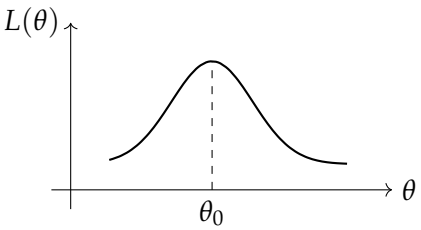
$$L_n(\theta) \xrightarrow{P} \mathbb{E}\big[\log f(X;\theta)\big] =: L(\theta) \text{ (true avg)}.$$

Fact 1. $L_n(\theta) \to L(\theta)$ as $n \to \infty$.

$L(\theta)$ is maximized at $\theta = \theta_0$:

$$\begin{aligned}
L(\theta_0) - L(\theta) &= \mathbb{E}_{\theta_0}\big[\log f(X;\theta_0) - \log f(X;\theta)\big] = \mathbb{E}_{\theta_0}\left[\log \frac{f(X;\theta_0)}{f(X;\theta)}\right]\\
&= D_{KL}(f_{\theta_0} \,\|\, f_{\theta}) \;\geq 0
\end{aligned}$$

(KL divergence ≥ 0 — 2টা distribution কতটুকু diverged). $\therefore\; L(\theta_0) \geq L(\theta) \Rightarrow \theta_0$-তে L function এর 1টা peak আছে।



Fact 2. $L(\theta)$ peaks at θ_0.

From Facts 1 & 2, as $n \to \infty$, MLE tends to θ_0. $\therefore$ MLE is consistent.

Caution. But $\geq$ only; we cannot ensure we have only *one* peak. So, one more condition: **identifiable** $\rightarrow$ only one peak. Identifiable হলে, 2টা func. এর value সমান মানে x-axis এও same হবে। $\therefore$ $L(\theta_0) > L(\theta)$ for $\theta \neq \theta_0$.

MLE is consistent if the model is identifiable.

Note. Function টা LLN change করছে, এর param-ও vary করছে — so LLN দিয়ে proof টা formal proof না। The formal version needs $\sup_\theta |L_n(\theta) - L(\theta)| \to 0$ (slide) — just mention করব যে এটা দিয়ে করা লাগে।

Lecture 7 Equivariance of the MLE; Hypothesis Testing 29/08/26

Equivariance

Let $\tau = g(\theta)$ and $\hat{\tau} = g(\hat{\theta}_n)$. $\mathcal{L}_n(\tau)$: likelihood func, যেটা τ input নেয় & data এর likelihood output দেয়।

$$\mathcal{L}_n^*(\tau)\text{-কে maximize করা} \;=\; \mathcal{L}_n(\theta) = \mathcal{L}_n\big(g^{-1}(\tau)\big)$$

$g^{-1}(\tau)$-কে $\hat{\theta}$ হতে হবে। $\therefore$ $\boxed{\hat{\tau} = g(\hat{\theta})}$.

Proof shown for invertible g; non-invertible $\rightarrow$ syllabus-এ নাই।

Note. CT syllabus $\rightarrow$ first 2 slides.

Hypothesis Test

- **Null hypothesis** (H_0): যে hypothesis টাকে test করব।
- **Alternative hypothesis** (H_1).

We cannot "accept" H_0. We can only **reject / not reject** it.

Research-এর claim এর উল্টা-টাকে null ধরি।

- reject $\rightarrow$ evidence for our claim;
- not reject $\rightarrow$ not sure.

Hypothesis test এর parameter: e.g. disease rate same in two groups $\rightarrow$ avg $= \mu$. So params can be $\mu, \theta, \ldots$

$\Theta \rightarrow$ set of possible values of θ. E.g. avg আয়ু $\rightarrow$ 65 yr:

$$H_0 : \theta \in \Theta_0 \; (= 65), \qquad H_1 : \theta \in \Theta_1 \; (\text{anything but } 65)$$

— the parameter space is divided.

একেকটা মানুষের individual আয়ু $\rightarrow$ data, X (r.v.).

$X \in R$ reject null
$X \notin R$ retain / not reject null

R: **rejection region**

Θ_1 and R are not the same: one is of the *param. space*, the other is of *data combinations*.

$$R = \{x : T(x) > c\}$$

T: **test statistic** function; c: **critical value**. যে input এর জন্য $T(x) > c$ হয়, ওটা included in the rejection region.

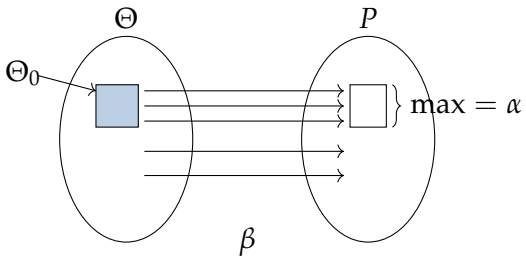
Type I and Type II Errors

- **Type I error:** ভালো মানুষকে punish করা — rejecting H_0 when H_0 is true.
- **Type II error:** খারাপ মানুষকে punish না করা — retaining H_0 when H_0 is false.

Note. Exam → definition + table (slide).

Definition (Power function). Random data rejection region-এ চলে যাওয়ার probability:

$$\beta(\theta) = P_\theta(X \in R).$$



Definition (Size / significance).

$$\alpha = \sup_{\theta \in \Theta_0} \beta(\theta)$$

Null এর পক্ষের θ গুলোর β value-এর maximum = max probability of Type I error.
Level α: upper bound of Type I error — test করার আগে থেকে given.

$\alpha \uparrow \Rightarrow$ rejection events $\uparrow$. 0 Type I error → আমাদের test reject-ই হবে না।

Simple vs. Composite; One- vs. Two-sided

- $\theta = \theta_0$ (e.g. গড় আয়ু 30) → **simple** hypothesis.
- $\theta > \theta_0$ or $\theta < \theta_0$ → **composite** hypothesis.

H_0:	$\theta = \theta_0$	$\theta \le \theta_0$	$\theta \ge \theta_0$
H_1:	$\theta \ne \theta_0$	$\theta > \theta_0$	$\theta < \theta_0$
	two-sided	one-sided	one-sided

Note. MLE chart → MLE derive, bias/unbiased, consistent.

Lecture 8 Power Function Example & the Wald Test 30/08/26

$R = \{x : T(x) > c\}$ — T শুরুতেই given; c reverse engineering করে বের করব।

Example (One-sided test for a Normal mean). $X_1, \ldots, X_n \sim N(\mu, \sigma^2)$, $T = \bar{X}$.

$$H_0 : \mu \le 0,\ \Theta_0 = (-\infty, 0]; \qquad H_1 : \mu > 0,\ \Theta_1 = (0, \infty).$$

Recall $\mathbb{E}[\bar{X}] = \mathbb{E}[X] = \mu$, $\text{Var}(\bar{X}) = \frac{\text{Var}(X)}{n} = \frac{\sigma^2}{n}$, so $\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$.

$$\beta(\mu) = P_\mu(\bar{X} > c) = P_\mu\left(\frac{(\bar{X}-\mu)\sqrt{n}}{\sigma} > \frac{(c-\mu)\sqrt{n}}{\sigma}\right) = P_\mu\left(\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} > \frac{(c-\mu)\sqrt{n}}{\sigma}\right)$$
$$= P\left(Z > \frac{\sqrt{n}(c-\mu)}{\sigma}\right) = 1 - \Phi\left(\frac{\sqrt{n}(c-\mu)}{\sigma}\right), \qquad Z \sim N(0,1)$$

($\bar{X}$-এর Z normalization: r.v. – r.v. এর গড়, divided by r.v. এর std. dev.; Φ = Normal এর CDF.)

α পেতে এই পুরোটা maximize করতে হবে $\Rightarrow$ CDF-কে minimize $\rightarrow$ CDF এর input-কে minimize: $\frac{\sqrt{n}(c-\mu)}{\sigma}$-কে minimize $\rightarrow$ μ-কে maximize. $\therefore$ $\mu \uparrow \Rightarrow \beta(\mu) \uparrow$. μ এর সর্বোচ্চ মান ∞, but we only look in $\Theta_0 = (-\infty, 0]$ $\therefore$ $\mu = 0$.

$$\alpha = \beta(0) = 1 - \Phi\left(\frac{\sqrt{n}\,c}{\sigma}\right) \implies c = \frac{\sigma\,\Phi^{-1}(1-\alpha)}{\sqrt{n}}$$

We don't use Φ^{-1} directly — reverse lookup z in the table.

The Wald Test

Wald test $\rightarrow$ **most important**. Scalar param: 1 dimension, 1 param.

Requirement. যত বেশি data collect করব, $\hat{\theta}$ asymptotically normal:

$$\frac{\hat{\theta} - \theta}{\widehat{\text{se}}} \rightsquigarrow N(0,1).$$

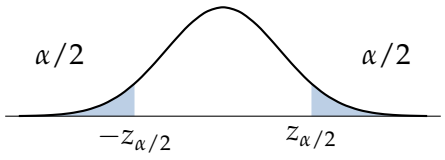
(Exam-এ Wald test আসলে verify এর দরকার নাই — direct test.)

Definition (Wald test). $H_0 : \theta = \theta_0$ vs. $H_1 : \theta \neq \theta_0$.

$$W = \frac{\hat{\theta} - \theta_0}{\widehat{\text{se}}}$$

(যদি H_0 true হয়, সেই estimate এর normalized z-value). **Reject H_0 when** $|W| > z_{\alpha/2}$ ($|W|$: test statistic; $z_{\alpha/2}$: critical value).

$z_{0.2}$: Normal dist.-এ কোন value এর জন্য তার থেকে বড় হওয়ার chance 0.2.



rejection region = both shaded tails

H_0 true ধরে, we retain when $-z_{\alpha/2} < W < z_{\alpha/2}$.

Proof: Wald test এর size α (asymptotically) — significance level α, how? (slide)

Comparing Two Proportions with the Wald Test

1টা program-এ কয়েকটা testcases আছে; some pass, some fail $\rightarrow$ Binomial. এরকম 2টা আলাদা আলাদা programs, diff. testcases.

$H_0 : p_1 = p_2$ ✗ — Wald test cannot work on 2 params. So let $\delta = p_1 - p_2$:

$$H_0 : \delta = 0, \qquad H_1 : \delta \neq 0.$$

MLE (from the Binomial MLEs): $\hat{\delta} = \hat{p}_1 - \hat{p}_2$.

$\hat{\theta}$ এমনভাবে guess করা উচিত, যেন $\widehat{\text{se}}$ easily guess করা যায় — not too simple either. Best way: MLE দিয়ে $\hat{\theta}$.

$$\begin{aligned}
\text{Var}(\hat{\delta}) &= \text{Var}(\hat{p}_1 - \hat{p}_2) = \text{Var}(\hat{p}_1) + \text{Var}(\hat{p}_2) \quad \text{(independent তাই)} \\
&= \text{Var}\left(\frac{X}{m}\right) + \text{Var}\left(\frac{Y}{n}\right) = \frac{\text{Var}(X)}{m^2} + \frac{\text{Var}(Y)}{n^2} \\
&= \frac{mp_1(1-p_1)}{m^2} + \frac{np_2(1-p_2)}{n^2} = \frac{p_1(1-p_1)}{m} + \frac{p_2(1-p_2)}{n} \\
&\therefore \ \widehat{\text{se}} = \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{m} + \frac{\hat{p}_2(1-\hat{p}_2)}{n}}
\end{aligned}$$

Caution. But একই testcase-এ 2টা program run দিলে এই independence থাকবে না।

Diff approach (paired data): $D_i = X_i - Y_i$.

$$\delta = \mathbb{E}[D_i] = \mathbb{E}[X_i - Y_i] = p_1 - p_2, \qquad \hat{\delta} = \bar{D} = n^{-1}\sum_i D_i, \qquad \widehat{\text{se}}(\hat{\delta}) = \frac{S}{\sqrt{n}}$$

(S = sample std. dev. of the D_i), then apply the Wald test.

Comparing 2 Means — Wald Test via a Confidence Interval

"Perform the Wald test using a $1-\alpha$ confidence interval":

1. $C = \left(\hat{\theta} - \widehat{\text{se}}\, z_{\alpha/2},\ \hat{\theta} + \widehat{\text{se}}\, z_{\alpha/2}\right)$ বানাব।
2. θ_0 টা confidence interval-এ আছে কিনা check করব: reject H_0 iff $\theta_0 \notin C$.

কিছু না বললে value / interval — any way.

Lecture 9 Statistical vs. Scientific Significance; p-values 31/08/26

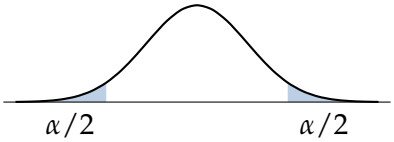
Definition (Statistically significant). Hypothesis test এর result যদি reject H_0 হয়।

- H_0 এর কাছাকাছি result এলেও reject;
- H_0 থেকে অনেক দূরে result এলেও reject → **scientifically significant** (more).

Example. C.I.: 60–70 yrs vs. C.I.: 40–90 yrs. যত বড় region, এই বড় range-এও ছিল না, তাই reject → more scientifically significant. But 64–66 → reject করলেও strong claim না।

Note. Exam: difference, definition.

$\alpha \uparrow\uparrow \Rightarrow$ rejection এর area বাড়ে $\Rightarrow$ reject করতে গিয়ে Type I error এর risk $\uparrow\uparrow$ (equivalently, $\alpha \downarrow\downarrow$ makes rejection harder).



p-value

$\alpha = 0.9$	rejected
0.8	rejected
0.7	rejected
0.6	rejected $\leftarrow$ α এর minimum যে value এর জন্য H_0 reject হয় = **p-value**
0.5999	not rejected

Definition (p-value).

$$\text{p-value} = \inf\{\alpha : X^n \in R_\alpha\}$$

(X^n: n sample এর data; R_α: rejection region at level α). Reject হওয়ার জন্য যেসব α দায়ী, তাদের infimum / minimum. **The p-value is data dependent.**

α আমরা tester-কে দেই। যখন tester-কে α দেয়া হয় না, p-value তখন applicable (Exam-এ α না বলা থাকলে):

- p-val $= 0.1$: $\alpha \geq 0.1$ হলে reject, নাহলে retain.
- p-val $\downarrow \Rightarrow$ H_0 এর against-এ evidence এর strongness $\uparrow\uparrow$.

Caution. p-value $\neq P(H_0 \mid \text{Data})$. A p-value of 0.9 doesn't mean H_0 gets "true-er". Hypothesis test can never decide truthfulness; it can only decide মিথ্যা কতটুকু।

Intuition. $n = 4$, my dataset $x^n = [10, 20, 30, 6]$ gives $T(x^n) = 10$ (test statistic এর input = dataset). For a random dataset X^n: $P\big(T(X^n) \geq 10\big)$ — random dataset এর result আমার dataset এর result এর চেয়ে বড়/equal হওয়ার probability. সব $\theta \in \Theta_0$ এর জন্য probability নিয়ে maximum (1টা থাকলে ওটাই).

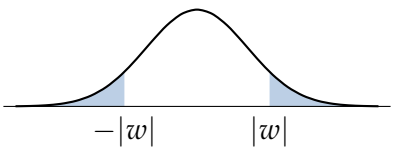
Via critical values. $\alpha \uparrow\uparrow \Rightarrow c_\alpha \downarrow\downarrow$. α বড় করতে করতে c_α crosses $|W|$ — সেই সময়ে α = p-value. So at the minimum α, $c_\alpha = |W|$ (critical val. = observed val.), where $t_{\text{obs}} = T(x^n)$.

$$\text{p-value} = \sup_{\theta \in \Theta_0} P_\theta\big(T(X^n) \geq c_\alpha\big) = \sup_{\theta \in \Theta_0} P_\theta\big(T(X^n) \geq T(x^n)\big)$$

(null এর পক্ষের params এর জন্য). Proof exam-এ আসতে পারে।

Wald test p-value:

$$\text{p-value} = P\big(|Z| > |w|\big) = 2\,\Phi(-|w|) \qquad (\Phi : \text{CDF})$$



p-value বের করা $\rightarrow$ always a safe choice. α দেয়া থাকলে compare করে conclude.

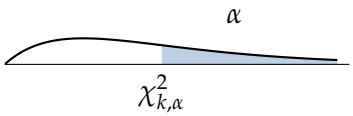
The χ^2 Test

Motivation. Decision tree: data-কে overfit করে। Overfitting এর solution: depth limited, অথবা পুরা depth পর্যন্ত বানানোর পর branch-এ গিয়ে χ^2-test করে prune.

Definition (χ^2 distribution).

$$V = \sum_{i=1}^{k} Z_i^2 \sim \chi^2_k \qquad (k = \text{d.o.f.})$$

$\chi^2_{k,\alpha}$: the point with right-tail area α (d.o.f. $= k$).



Pearson's χ^2 Test for Multinomial Data

$$X = (X_1, X_2, \ldots, X_k) \sim \text{Multinomial}(n, p), \qquad p = (p_1, \ldots, p_k)$$

(X_1: dice-এ 1 কয়বার উঠেছে; k categories). Question: underlying tuple p অন্য কোন tuple p_0 এর সাথে same কিনা।

Caution. The Wald test is not applicable for this (it's a whole tuple, not a scalar).

Dice thrown n times: 2 উঠবে $\to$ $n\,p_{02}$ times; j উঠবে $\to$ $n\,p_{0j}$ times (*theoretical*). আমার observation-এ j উঠেছে $\to$ X_j times (*observed*).

$$T = \sum_{j=1}^{k} \frac{(X_j - np_{0j})^2}{np_{0j}} \qquad (np_{0j} = \mathbb{E}[X_j] \text{ under } H_0)$$

Reject H_0 when $T > \chi^2_{k-1,\alpha}$; p-value $= P(\chi^2_{k-1} > t)$.

Example (Mendel's peas — 4 types)**.** Claim: $p_0 = \left(\frac{9}{16}, \frac{3}{16}, \frac{3}{16}, \frac{1}{16}\right)$. $n = 556$ trials, observed $X = (315, 101, 108, 32)$.

$$T = \cdots = 0.47$$

[α not given] $\Rightarrow$ p-val $= P(\chi^2_3 > 0.47) = 0.93$ (table-এ CDF খুঁজে $1 - \text{CDF}$). $\Rightarrow$ H_0 এর against-এ weak evidence.

Permutation Test

scalar check	$\to$ Wald
tuple check	$\to \chi^2$
distribution check	$\to$ permutation test

$X_1, \ldots, X_m \sim F_X, \quad Y_1, \ldots, Y_n \sim F_Y.$

$$H_0 : F_X = F_Y, \qquad H_1 : F_X \neq F_Y$$

$$T(X_1, \ldots, X_m, Y_1, \ldots, Y_n) = \left|\bar{X}_m - \bar{Y}_n\right|$$

($\bar{X}_m$: 1ম m জনের গড়, $\bar{Y}_n$: পরের n জনের গড়; T X, Y চিনে না — it only sees positions). $(m+n)! = N!$ shuffles ($m + n = N$). Out of these $N!$ shuffles, কয়টায় আমার চেয়ে larger val. পায়, divided by $N!$ = prob.:

$$\text{p-value} = P(T > t_{\text{obs}}).$$

Example. $T(X_1, X_2, Y_1) = |\bar{X} - \bar{Y}| = 2$. $3! = 6$ shuffles; কয়টায় > 2? $= 4$টা. $\therefore$ p-val $= 4/6$.

Disadvantage: অনেক বড় set হলে সব shuffle try করা tough.

Lecture 10 Multiple Testing: Bonferroni & Benjamini–Hochberg 05/09/26

Multiple Testing Problem

Combine all hypothesis tests, যেটার overall α যেন (given level) থাকে। p-values: $P_1, P_2, \ldots, P_m$.

Note. Recall: p-value = min α for rejection; $\alpha >$ p-value $\Rightarrow$ reject.

Definition (Bonferroni method). Reject test i if $P_i < \dfrac{\alpha}{m}$ (individual test গুলোর α). এভাবে assign করলে max Type I error = α হবে।

1টা H_0 (falsely) reject করলেই Type I error. Let R: at least 1টা test-এ H_0 falsely rejected.

$$P(R) = P\left(\bigcup_{i=1}^{m} R_i\right) \leq \sum_{i=1}^{m} P(R_i) = \sum_{i=1}^{m} \frac{\alpha}{m} = \alpha$$

($\leq$, not $=$: R_i গুলো independent হতে পারে but not mutually exclusive events.)

Example. $\alpha = 0.05$, $m = 2638$ genes $\rightarrow \frac{\alpha}{m} = 0.00001895375$ — at most Type I error কতটুকু tolerate করব। বেশি ছোট হয়ে গেলে test reject-ই করতে পারে না। Type I error কমলেও rejection-ও কমে যায়। [Not used much]

False Discovery Rate

	H_0 not rejected	H_0 rejected
H_0 true	U	V
H_0 false	T	S

V: null true হলেও got rejected; $R = V + S$: total rejections.

Definition (FDP and FDR). FDP = total rejections এর কত proportion falsely rejected:

$$\text{FDP} = \begin{cases} V/R, & R > 0 \\ 0, & R = 0 \end{cases}$$

FDP is *not fixed* — random, কারণ data random. On avg FDP কত পাই $\rightarrow$ **FDR** (constant):

$$\text{FDR} = \mathbb{E}(\text{FDP}).$$

Benjamini–Hochberg (BH) Method

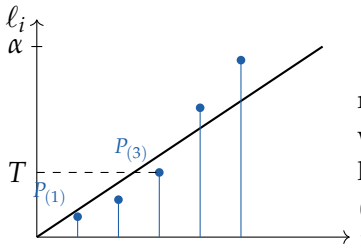
m সংখ্যক hypothesis test, প্রত্যেকের p-value; sort করব:

$$P_{(1)} < \cdots < P_{(m)}.$$

$$\ell_i = \frac{i\,\alpha}{C_m\, m} \qquad (C_m = \text{correction factor}), \qquad \text{e.g. } \ell_1 = \frac{\alpha}{m},\ \ell_2 = \frac{2\alpha}{m},\ \ell_3 = \frac{3\alpha}{m}\ (C_m = 1)$$

এমন একটা index choose করব, যে এই line এর নিচে আছে — ওগুলোর last:

$$R = \max\{i : P_{(i)} < \ell_i\}, \qquad T = P_{(R)} \text{ is the rejection threshold.}$$



rightmost index whose $P_{(i)}$ lies below the line $= R$ (here $R = 3$)

Reject all H_0 with $P_{(i)} \leq T$.

FDR $\leq \alpha$ (proof syllabus-এ নাই).

$$C_m = 1 \ \text{(tests independent) [default]}; \qquad C_m = \sum_{i=1}^{m} \frac{1}{i} \ \text{(not independent)}.$$

Bonferroni bounds the overall test's $\alpha \rightarrow$ each individual test gets too small an α; **BH resolves it**.

Note. BH math $\rightarrow$ TF.

Example (Worked procedure — TF)**.** p-values: __, __, ... (10টা); overall $\alpha = 0.05$, $m = 10$.

- **Bonferroni:** $\frac{\alpha}{m} = 0.005 \rightarrow$ compare with each P_i; $\frac{\alpha}{m} > P_i$ হলে reject.
- **BH:** sort p-values; find the ℓ_j array; check $P_{(1)} < \ell_1$, $P_{(2)} < \ell_2, \ldots, P_{(j)} < \ell_j$ কিনা। Take the rightmost such i; its $P_{(i)}$ becomes the effective threshold. E.g. $P_{(1)}, \ldots, P_{(5)} < \ell_1, \ldots, \ell_5$, $P_{(6)} \not< \ell_6 \Rightarrow$ reject $P_{(1)}, \ldots, P_{(5)}$.

BH বেশি reject করতে পারল।

- Bonferroni: conservative, $\alpha = 0.005$ — narrow space for error.
- BH: wider, ≈ 0.02 — বেশি α.

Bayesian Inference

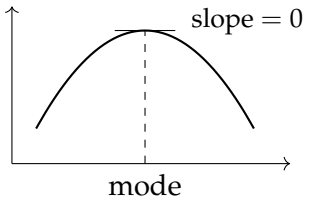
Frequentist (purist)	**Bayesian**
10টা coin flip $\rightarrow$ 2 জন আলাদা করল। 2 জনের data আলাদা, random. But the parameter is true, fixed; জানিনা দেখে random এমন না।	আমি যা জানি না, সেটা আমার জন্য random. 90% C.I.: param lie করার chance 90% — আগে wrong হলেও Bayesian is fine with it.

Bayesian Method

Underlying Bernoulli p parameter একটা **"prior"** distribution define করব। কিছু জানি না $\rightarrow$ Uniform dist., $f(\theta) \rightarrow$ constant. Then update

$$f(x \mid \theta) \longrightarrow f(\theta \mid X_1, \ldots, X_n) \qquad \text{(details: slide)}.$$

- Bayes $\rightarrow$ **posterior mean**: $\bar{\theta}_n = \int \theta \, f(\theta \mid x^n) \, d\theta$ (value $\times$ probability).
- Freq. $\rightarrow$ MLE.
- **MAP:** posterior এর mean না নিয়ে, **mode** নিব (চরম মান $\Rightarrow$ slope $= 0$).



MAP vs MLE (Exam).

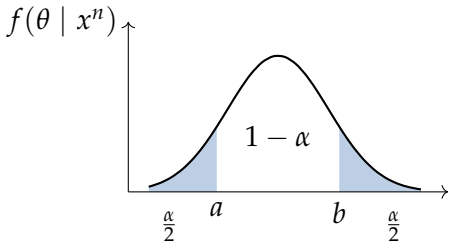
$$\hat{\theta}_{\text{MAP}} = \arg\max_{\theta} \big(\log \mathcal{L}(\theta) + \log f(\theta) \big)$$

MAP considers the prior; MLE doesn't consider the prior (বা consider করে, but সবগুলো θ-কে equally likely ধরে — শুধু data দেখে maximize). $f(\theta)$ indep./uniform/const. হলে, **MAP = MLE**.

Example. 100টা toss: 80 H, 20 T $\to$ MLE: does not consider the prior at all. Bayesian: data না দেখেই idea নেয় (fair coin); both prior and after-data-কে maximize করে।

Lecture 11 Posterior Intervals, Beta Prior & Laplace's Rule 06/09/26

CI: Frequentist; Bayesian-এ বলি **"Posterior Confidence Interval"**.



এমন 2টা value a, b choose করব: a এর left-এ $\frac{\alpha}{2}$, b এর right-এ $\frac{\alpha}{2}$. Param. এই interval-এ belong করার prob. $= 1-\alpha$.

$$\underbrace{P_\theta(\theta \in C_n) = 1-\alpha}_{\text{frequentist}} \qquad \text{vs.} \qquad \underbrace{P(\theta \in C \mid x^n) = 1-\alpha}_{\text{Bayesian (given the data)}}$$

Frequentist	**Bayesian**
C_n is random, θ is fixed.	C is fixed, θ is random.
Repeated experiments-এ 95% interval can capture θ.	Particular data দেখে ওটার জন্য fixed posterior interval বানাই।
Data দেখে interval; data not given.	Data given, 1টাই experiment.

Note. Exam-এ ONLY mathematical notation না দিয়ে, বুঝিয়েও লিখতে হবে।

Beta Distribution

Beta(α, β) — 2টা param.

$$f(x;\alpha,\beta) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}\, x^{\alpha-1}(1-x)^{\beta-1}, \qquad 0 < x < 1$$

success$^{\alpha-1}$ failure$^{\beta-1}$ দেখলে বুঝব β-dist: β(success এর power $+1$, failure এর power $+1$).

- $\alpha = \beta = 1$: x term থাকে না — becomes Uniform. **Unif is a special type of β-dist.**
- Mean: $\mathbb{E}[X] = \dfrac{\alpha}{\alpha+\beta}$.

Bernoulli with a Uniform Prior

$X_1, \ldots, X_n \sim \text{Bernoulli}(p)$, uniform prior $f(p) = 1$. Frequentist: estimates with MLE; Bayesian: mean, mode (MAP), ...

$$f(p \mid x^n) \propto \underbrace{f(p)}_{\text{prior}} \times \underbrace{\mathcal{L}_n(p)}_{\text{likelihood}} = 1 \cdot p^s(1-p)^{n-s} \to \beta\text{-dist}$$

(n বার coin toss-এ s সংখ্যক H উঠে, $s = \sum_{i=1}^n X_i$.) Exact sequence-এর order already বের করছি; no need for a $\binom{n}{s}$-type coefficient — $\propto$ থাকায়ও লাগতো না।

$$p \mid x^n \sim \text{Beta}(s+1,\ n-s+1)$$

Bernoulli dist.-এর underlying parameter এর distribution, under a uniform prior. Posterior mean:

$$\bar{p} = \frac{s+1}{n+2} \qquad \text{vs. MLE} \quad \hat{p} = \frac{s}{n}.$$

"Laplace Succession Rule" → each category-তে 1টা করে value insert.

Example. H H H T T: naively $\hat{p} = \frac{3}{5}$; Bayesian $\bar{p} = \frac{3+1}{5+2} = \frac{4}{7}$ — naive এমন result পেতে add one H, one T.

Issue of MLE. MLE → 5 বার toss করে 5টা H: $\hat{p} = 1$, claiming that T is impossible. Laplace resolves this — কোন category-তে data না আসলেও count করা।

Uniform prior $\text{Unif}(0,1)$ এর mean $= \frac{1}{2} = \tilde{p}$. The posterior mean is a weighted mix:

$$\bar{p} = \lambda_n \hat{p} + (1-\lambda_n)\tilde{p}.$$

Bayesian estimate-কে linear comb. of MLE & prior দেখাতে গেলে,

$$\lambda_n = \frac{n}{n+2}; \qquad n \to \infty \text{ [অনেকবার coin toss] হলে } \lambda_n \approx 1.$$

$\lambda_n \to 1$ হলে, $\bar{p}$ & $\hat{p}$ তত close হবে (অনেক samples নিলে শুধু H আসার chance কমে)। λ_n যত 1 থেকে দূরে, prior এর প্রভাব তত বেশি।

Estimate w.r.t. posterior mode. $f(p \mid x^n) = p^s(1-p)^{n-s} \xrightarrow{\text{diff.}}$

$$\hat{p}_{\text{MAP}} = \frac{s}{n} = \hat{p}_{\text{MLE}}$$

(MAP asks for the prior, MLE doesn't — same here কারণ prior জানি না, i.e. uniform).

General Beta Prior

Let prior $p \sim \text{Beta}(\alpha, \beta)$:

$$f(p \mid x^n) \propto p^{\alpha-1}(1-p)^{\beta-1} \cdot p^s(1-p)^{n-s} = p^{(\alpha+s)-1}(1-p)^{(\beta+n-s)-1}$$

$$p \mid x^n \sim \text{Beta}(\alpha+s,\ \beta+n-s), \qquad \bar{p} = \frac{\alpha+s}{\alpha+\beta+n}$$

আগের value এর generic version [$\alpha = \beta = 1$ বসালে আগের value].

$$\bar{p} = \underbrace{\left(\frac{n}{\alpha+\beta+n}\right)}_{n\to\infty:\ \approx 1\ (\approx\text{MLE})} \hat{p} + \left(\frac{\alpha+\beta}{\alpha+\beta+n}\right) p_0, \qquad p_0 = \text{prior mean} = \frac{\alpha}{\alpha+\beta}.$$

Note. $\hat{p}_{\text{MAP}} \neq \hat{p}_{\text{MLE}}$ here. Why? Prior আর 1 (uniform) না। (Details.)

Prior vs. Posterior; Conjugate Priors

Prior	Posterior
data না দেখেই param এর prob.	data দেখে param এর prob.

Definition (Conjugate prior). Prior, posterior একই family means guessed distribution change হয় নি — overall, হয়তো small changes. এমন prior-কে বলি **"Conjugate Priors"**. (Definition বুঝিয়ে বলতে হবে।)

Example (Normal prior for a Normal mean). $X_1, \dots, X_n \sim N(\theta, \sigma^2)$, prior $\theta \sim N(a, b^2) \Rightarrow$ posterior $\theta \mid x^n \sim N(\bar{\theta}, \tau^2)$. So this prior is a conjugate prior.

95% interval $C = (c, d)$: $P(\theta < c \mid x^n) = P(\theta > d \mid x^n) = 0.025$.

$$P\left(\frac{\theta - \bar{\theta}}{\tau} < \frac{c - \bar{\theta}}{\tau} \,\middle|\, x^n\right) = P\left(Z < \frac{c - \bar{\theta}}{\tau}\right) = 0.025 \;\Rightarrow\; \frac{c - \bar{\theta}}{\tau} = -1.96 \;\Rightarrow\; c = \bar{\theta} - 1.96\tau$$

($\frac{\theta - \bar{\theta}}{\tau}$ is θ এর জন্য std. normal var.)

$$\therefore\ \bar{\theta} \pm 1.96\,\tau \ \text{(Bayesian)} \qquad \text{vs.} \qquad \hat{\theta} \pm 1.96\ \text{se} \ \text{(frequentist)}$$

Prior অন্য কিছু (Unif না) হলে এত মিল থাকত না।

Functions of Parameters

- **Frequentist:** $\text{MLE}(f(\theta)) = f(\text{MLE}(\theta))$ (equivariance).
- **Bayesian:** $\tau = g(\theta)$ — τ এর posterior dist. লাগবে $\to h(\tau \mid x^n)$.

Find the CDF first:

$$H(\tau \mid x^n) = P\big(g(\theta) \le \tau \mid x^n\big) = \int_A f(\theta \mid x^n)\, d\theta, \qquad A = \{\theta : g(\theta) \le \tau\}$$

(A: এমন সব θ যারা এই শর্ত follow করে), then differentiate $\to h(\tau \mid x^n)$.

Lecture 12 Posterior of a Transformed Parameter & Simulation 07/09/26

Example (Log-odds). $X_1, \dots, X_n \sim \text{Bernoulli}(p)$. Target: $\psi = \log\left(\frac{p}{1-p}\right)$ এর posterior $h(\psi \mid x^n)$. (Frequentist-এ: MLE এর equivariance theorem.) 40% H, 60% T: 4 to 6 odds. $\psi = \log(\text{odds}) =$ **log-odds (logits)**.

$$H(\psi \mid x^n) = P(\Psi \le \psi \mid x^n) = P\left(\log\frac{p}{1-p} \le \psi \,\middle|\, x^n\right) = P\left(p \le \frac{e^\psi}{1+e^\psi} \,\middle|\, x^n\right) \quad (p \text{ এর CDF})$$

$$= \int_0^{e^\psi/(1+e^\psi)} f(p \mid x^n)\, dp \quad (p \text{ এর PDF; upper limit-এ } \psi \text{ এর term})$$

Differentiate w.r.t. ψ, using

$$\frac{d}{dt}\int_a^{g(t)} f(x)\, dx = \frac{d}{dt}\Big[F(x)\Big]_a^{g(t)} = \frac{d}{dt}\big[F(g(t)) - F(a)\big] = f(g(t))\, g'(t)$$

(upper limit-কে func-এ বসিয়ে উপরের diff. দিয়ে গুণ):

$$h(\psi \mid x^n) \;\propto\; \left(\frac{e^\psi}{1+e^\psi}\right)^s \left(\frac{1}{1+e^\psi}\right)^{n-s} \times \frac{d}{d\psi}\left(\frac{e^\psi}{1+e^\psi}\right) = \cdots$$

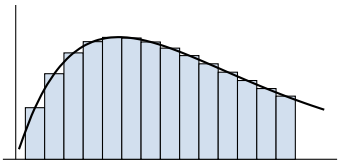
Then guess করব posterior এর mean/mode দিয়ে — usual way-তে ঝামেলা।

Simulation

Another way: simulation [যদি $p(\theta \mid x^n)$ জানি]:

$$\theta_1, \theta_2, \ldots, \theta_B \sim p(\theta \mid x^n)$$

Sample করে অনেকগুলো samples এনে ওগুলোর histogram (samples are discrete) আঁকলে $p(\theta \mid x^n)$-কে approximate করে। Posterior এর mean, mode বের না করে samples এর mean, mode বের করব।



- **Quartile:** split into 25% | 25% | 25% | 25%.
- **Quantile:** x%.

একপাশে $\frac{\alpha}{2}$%, আরেকপাশে $\frac{\alpha}{2}$% portion বাদ দিলে ঐ θ গুলোর interval-ই **posterior interval**.

Sample গুলা দিয়ে histogram না এঁকে individual samples-কে function এর ভিতর ঢুকালে ঐ function এর posterior-কে approximate করে (e.g. samples $\theta_b \mapsto \psi_b = \log \frac{\theta_b}{1-\theta_b}$, then histogram of ψ_b).

Important. Simulation গুলো নিজেরা করে interval + explain "why". E.g. samples given $(30, 10, \ldots)$ $\to$ logit $\to$ $\log(\cdots)$ $\to$ histogram.

Priors

Prior: data না দেখেই subjective guess, কাউকে কাউকে favor করে $\to$ not good.

Non-informative prior: coin যেহেতু toss করিনি, coin এর p $0.1, 0.2, \ldots$ anything হতে পারে $\to$ **flat prior**: uniform dist. (PDF এর value কত হবে?) যে prior গুলা equally likely বুঝাতে $f(p) = 1$ বোঝায়, but on an unbounded space density func. valid না (area $= \infty$?) — still works, যেহেতু $f(\theta \mid x^n) \propto \mathcal{L}_n(\theta) f(\theta)$ (proportional to, তাই issue না).

Issue of flat prior. Want to draw a square (length 10–20). Length: Uniform from 10 to 20 $\Rightarrow$ Area: হওয়া উচিত Uniform 100 to 400. But area কোন transformation এর পর flat আসলো না। "Reparametrize" করার কারণে distribution change হয়ে যায় $\to$ **invariant না**.
Flat prior is not invariant (in general): $f(p) = 1$ but after reparametrizing to ψ, $f(\psi) \neq 1$.

Change of Variables.

$$f_\Psi(\psi) = f_p(p) \left| \frac{dp}{d\psi} \right| \qquad \Longleftrightarrow \qquad f(\psi)\, d\psi = f(p)\, dp$$

Density func. $\times$ দৈর্ঘ্য $= \psi$ এর domain-এ কিছু ঘটনা. E.g. $z = x^2 \Rightarrow dz = 2x\, dx$: diff. domain-এ একই ঘটনা।

Note. Flat prior, reparametrization given $\to$ invariant? (Important.)

Flat prior: $\to$ non-informative, but $\to$ invariant না।

Jeffreys' Prior (Invariant & Non-informative)

$$f(\theta) \propto I(\theta)^{1/2}, \qquad I(\theta) = -\mathbb{E}_\theta\left[\frac{\partial^2}{\partial\theta^2}\log f(X;\theta)\right] \quad \text{(Fisher information; 2 বার diff)}$$

Example (Bernoulli). $I(p) = p^{-1}(1-p)^{-1}$, so

$$f(p) \propto \sqrt{I(p)} = p^{-1/2}(1-p)^{-1/2} \;\to\; \text{Beta}\left(\tfrac{1}{2},\tfrac{1}{2}\right).$$

For this prior: logit এর জন্য reparametrize করে check if invariant.

Multiparameter Problems

$\theta = (\theta_1, \ldots, \theta_p)$, $\quad f(\theta \mid x^n) \propto \mathcal{L}_n(\theta) f(\theta)$ (joint in $\theta_1, \theta_2, \ldots, \theta_p$). To focus on θ_1 alone we need its marginal — বাকি params integrate out করতে হয় → **hectic**.

Note. Exam-এ analytic way করতে দিবে না — just way আসতে পারে; calc. দিলে simulation চাবে।

Simulation: $\theta^1, \ldots, \theta^B \sim f(\theta \mid x^n)$ (B বার sample করলাম; একেকটা list/tuple). We want to focus on θ_1: প্রত্যেকটা list-এর প্রথম value নিয়ে (B সংখ্যক samples) histogram → mode, mean → posterior interval.

Example (Two binomials). $X_1 \sim \text{Binomial}(n_1, p_1)$, $X_2 \sim \text{Bin}(n_2, p_2)$, $f(p_1, p_2) = 1$. Posterior:

$$f(p_1, p_2 \mid x_1, x_2) \;\propto\; \underbrace{p_1^{x_1}(1-p_1)^{n_1-x_1}}_{f(p_1|x_1)}\;\underbrace{p_2^{x_2}(1-p_2)^{n_2-x_2}}_{f(p_2|x_2)}$$

(indep. দেখে আলাদা লেখা যাবে). Simulating:

$$P_{1,1}, P_{1,2}, \ldots, P_{1,B} \sim f(p_1 \mid x_1), \qquad P_{2,1}, P_{2,2}, \ldots, P_{2,B} \sim f(p_2 \mid x_2),$$

then use $\delta_b = P_{2,b} - P_{1,b}$ (or any function) for its posterior.

Bayesian Testing

Data থেকে null এর probability: $P(H_0 \mid x^n)$. $\approx 99\%$ হলে, for this data H_0 can be **accepted** (frequentist test-এ শুধু reject / not reject ছিল).

With $P(H_0) = P(H_1) = \frac{1}{2}$:

$$P(H_0 \mid X^n = x^n) = \frac{f(x^n \mid H_0)P(H_0)}{P(x^n)} = \frac{f(x^n \mid H_0)P(H_0)}{f(x^n \mid H_0)P(H_0) + f(x^n \mid H_1)P(H_1)}$$
$$= \frac{\frac{1}{2}f(x^n \mid \theta_0)}{\frac{1}{2}f(x^n \mid \theta_0) + \frac{1}{2}\int f(x^n \mid \theta) f(\theta)\, d\theta} = \frac{\mathcal{L}(\theta_0)}{\mathcal{L}(\theta_0) + \int \mathcal{L}(\theta) f(\theta)\, d\theta}$$

(θ_0: H_0 এর পক্ষের param; integral over θ_0 বাদে বাকি সব θ — 1টা particular value θ_0 বাদ দেয়া/না দিয়ে integral same.)

$\int \mathcal{L}(\theta) f(\theta)\, d\theta = \mathbb{E}[\mathcal{L}(\theta)]$ (avg). Avg-কে সরিয়ে সর্বোচ্চ মান (MLE) বসালে value $\downarrow\downarrow \Rightarrow$ পুরা func. এর **lower bound**:

$$P(H_0 \mid x^n) \;\geq\; \frac{\mathcal{L}(\theta_0)}{\mathcal{L}(\theta_0) + \mathcal{L}(\hat{\theta})}.$$

Lower bound বড় ($\approx 90\%$) → actual val. আরো বেশি হবে, so accept. Lower bound $\approx 5\%$ → কিছু বলা যায় না।

Lecture 13 Markov Chains: Basics & Chapman–Kolmogorov 12/09/26

Book: Sheldon Ross (exercises); Rec.: MDAA.

X_1: rains today, X_2: rains tomorrow, X_3: ... (X_1, X_2, X_3: time index). Predict the future without knowing the full past: X_n জেনে X_{n+1} model করা।

$\{X_n,\ n = 0, 1, 2, \dots\}$: n = time stamp এর index, X_n = state; $X_n = i$: time n-এ X i-state এ আছে।

Definition (Markov property). nth TS-এর state জানলেই $(n+1)$th TS-এর state জানব — full history জানা লাগে না:

$$P\{X_{n+1} = j \mid X_n = i,\ \underbrace{X_{n-1} = i_{n-1}, \dots, X_0 = i_0}_{\text{matter করে না}}\} = P_{ij}$$

P_{ij}: 1 TS পর i থেকে j-তে যাওয়ার probability.

Problem-টা Markov property মানলেই Markov chain হবে।

$$\sum_{j=0}^{\infty} P_{ij} = 1 \qquad \text{(intuition: কোথাও না কোথাও যাবেই; } i \text{ itself included).}$$

Transition Matrix (single-step)

$$\mathbf{P} = \begin{pmatrix} P_{00} & P_{01} & P_{02} & \cdots \\ P_{10} & P_{11} & P_{12} & \cdots \\ \vdots & \vdots & \vdots & \\ P_{i0} & P_{i1} & P_{i2} & \cdots \\ \vdots & \vdots & \vdots & \end{pmatrix}$$

Row: আজকে কোন state-এ (given); column: future-এ কোন state-এ যাব। $\sum_{1 \text{ row}} P = 1$.

Example (Rain). States: $0 =$ rain, $1 =$ no rain. $P(\text{rains tomorrow} \mid \text{rains today}) = \alpha$, $P(\text{rains tomorrow} \mid \text{doesn't rain today}) = \beta$.

Today\Tomorrow	R	N
R	α	$1-\alpha$
N	β	$1-\beta$

যা দেয়া আছে, সেগুলো always row-তে বসবে এমন না। (R/N labels exam-এ লেখা দরকার নাই।)

Example (Binary bit). A binary bit under a transformation: changes w.p. p, unchanged w.p. $1-p$. States $\{0, 1\}$:

$$\mathbf{P} = \begin{pmatrix} p & 1-p \\ 1-p & p \end{pmatrix} \quad \text{(rows/cols } 0, 1\text{)}$$

Check: this matrix matches "stays the same w.p. p". If p is the probability of *change*, the diagonal should be $1-p$ and the off-diagonal p.

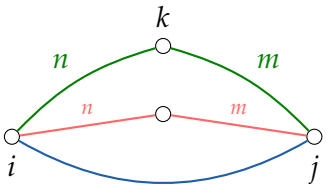
n-step Transition Probabilities

$$P_{ij}^n = P\{X_{n+k} = j \mid X_k = i\}, \qquad P_{ij}^1 = P_{ij}$$

কোন TS k-তে i state-এ ছিলো; nটা step পরে j state-এ। Why not a function of k? — Markov property.

Chapman–Kolmogorov Equation (Floyd–Warshall এর মতো).

$$P_{ij}^{n+m} = \sum_{k=0}^{\infty} P_{ik}^{n} P_{kj}^{m}$$



along a path: AND (multiply)

across paths: OR (add)

Formal proof (may appear on exam):

$$\begin{aligned} P_{ij}^{n+m} &= P\{X_{n+m} = j \mid X_0 = i\} = \sum_{k=0}^{\infty} P\{X_{n+m} = j,\ X_n = k \mid X_0 = i\} \\ &= \sum_{k} P\{X_{n+m} = j \mid X_n = k,\ X_0 = i\}\, P\{X_n = k \mid X_0 = i\} = \sum_{k} P_{kj}^{m}\, P_{ik}^{n}. \end{aligned}$$

- শুরুতে X_0, X_1 নাকি X_{100} matter করে না। $X_2 = i$ লিখলে $P\{X_{2+n+m} = j \mid X_2 = i\}$ লিখতে হতো।
- "Sum out": $A \cap B$, $A \cap \bar{B}$ মিলে শুধু A থাকে।
- P_{kj}^{m}: 1টা mat. এর row idx k; P_{ik}^{n}: অন্য mat. এর column idx k $\Rightarrow$ matrix multiplication:

$$\mathbf{P}^{(n+m)} = \mathbf{P}^{(n)} \cdot \mathbf{P}^{(m)}, \qquad \mathbf{P}^{(4)} = \left(\mathbf{P}^{(2)}\right)^2, \qquad \mathbf{P}^{11} \Rightarrow 11 = 1011_2 \text{ (fast exponentiation)}.$$

$\mathbf{P}$ এর power বাড়ালে আজকেরটা fixed; future TS-টা shift হয়।

	Sat	Sun	Mon
$\mathbf{P}^1$	today	future	
$\mathbf{P}^2$	today		future

Example. (Rain chain with $\alpha = 0.7, \beta = 0.4$.) $\mathbf{P}^4 = \begin{pmatrix} 0.5749 & 0.4251 \\ \cdots & \cdots \end{pmatrix}$ — 0th row: today rained; 1st col: 4 দিন পর didn't rain.

Note. A distribution over today's states C (row vector, e.g. $[0.7\ \ 0.3]$): n দিন পর $F = C\,\mathbf{P}^n$. Going back: $C = F(\mathbf{P}^n)^{-1}$ — but $\mathbf{P}$ may be non-invertible $\rightarrow$ multiple possible previous states.

Accessibility & Communication

Definition (Accessible). কোন না কোন TS-এ i থেকে j-এ যাওয়া possible: $P_{ij}^{n} > 0$ for some $n \geq 0$. ($n = 0$: state 5 is accessible from state 5.) $i \leftrightarrow j$: i, j communicate.

Transitivity: if $i \leftrightarrow j$ and $j \leftrightarrow k$, then $i \leftrightarrow k$:

$$P_{ik}^{n+m} = \sum_{r=0}^{\infty} P_{ir}^{n} P_{rk}^{m} \ \geq\ P_{ij}^{n} P_{jk}^{m} > 0$$

(loop-এ at some point $r = j$ হবে, so sum ঐ term থেকে $\geq$ হবে).

Lecture 14 Irreducibility, State Augmentation, Limiting Probabilities 13/09/26

Definition (Irreducible Markov chain). All pairs of states communicate with each other.

$$\begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ 0 & \frac{1}{3} & \frac{2}{3} \end{pmatrix}$$

Transition matrix দিয়ে irreducible কিনা (important). কোন 0 entry নাই $\Rightarrow$ irreducible. 0 entry থাকলে ওগুলোকে double-check (multi-step path আছে কিনা) — the matrix above is irreducible ($0 \to 1 \to 2$, $2 \to 1 \to 0$).

Making a Process Markov (state mapping — tough)

জোর করে Markov property hold করে এমনভাবেই model বানাব। গত 2 দিনের উপর current state depend করলে, make a state combining both: $00, 01, 10, 11$ — 4 রকম state.

Example. Given: $P(\text{rain tomorrow} \mid \text{rain yesterday \& today}) = 0.7$, $P(\cdot \mid \text{N yesterday, R today}) = 0.5$, ... (table in question).

Solution: future state হবে (আজকে + tomorrow). Present state = (Yesterday, Today), next state = (Today, Tomorrow).

$$0 : \text{RR}, \quad 1 : \text{NR}, \quad 2 : \text{RN}, \quad 3 : \text{NN} \qquad (\text{day1, day2})$$

R R $\to$ R: $P_{00} = 0.7$; N R $\to$ R: $P_{10} = 0.5$; ... এভাবে 8টা entries fill up. Matrix-এ ঘর 16টা; বাকি 8টা entry 0 — e.g. P_{01} is RR$\to$NR: impossible transition (today must stay R).

Q: গতকালকেও বৃষ্টি, আজকেও বৃষ্টি, পরশু বৃষ্টি? $P = ? \Rightarrow$ find $\mathbf{P}^2$. (Yesterday, Today) R R given; Tomorrow N/R — we don't care; Day after tomorrow R — what we target. $\therefore$ 2টা transition এর OR করব: the final state (Tomorrow, Day after) must end in R, i.e. RR (0) or NR (1):

$$\text{ans} = P_{00}^2 + P_{01}^2.$$

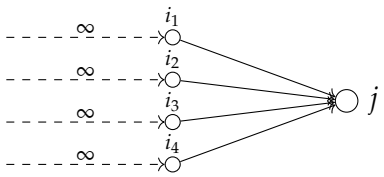
Limiting Probabilities

$$\lim_{n \to \infty} P_{ij}^n = \pi_j$$

Infinite steps-এ i থেকে j-তে যাওয়ার probability (i যায় আসে না, কিছু একটা হলেই হবে).

$$\pi_j = \sum_i \pi_i P_{ij}$$

∞ steps দিয়ে কোন i-তে এসে, তারপর 1 step-এ j. i-তে না গিয়ে k এর through-ও যেতে পারে; one path contributes to π_j — সব sum করলে π_j পাব।



Stationary equations.

$$\pi_j = \sum_i \pi_i P_{ij}, \qquad \sum_j \pi_j = 1 \qquad \Longleftrightarrow \qquad \pi_{1\times n} = \pi_{1\times n}\,\mathbf{P}_{n\times n}$$

(π-কে index করছি শুধু j দিয়ে, তারমানে π is a row matrix.)

For 2 states:

$$\begin{bmatrix}\pi_0 & \pi_1\end{bmatrix} = \begin{bmatrix}\pi_0 & \pi_1\end{bmatrix}\begin{bmatrix}P_{00} & P_{01}\\ P_{10} & P_{11}\end{bmatrix}$$

$\Rightarrow$ 2টা equation পাব; solve করলে π_0, π_1 পাব (slide).

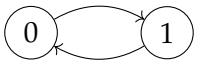
Example. Rain chain: $\pi_0 = 0.571 = 57.1\%$ (rain). কাল/পরশু wise chance না, কারণ **P** এর power কত নিব বুঝব না। π can tell us the probability on a random day $\rightarrow$ **"Long Run Probability"**.

Typical workflow. 1. Form **P** 2. Predict 3. Limiting probability

Periodicity

Definition (Period). $P_{ii}^n = 0$ যদি steps d দ্বারা divisible না হয়, d is the largest such integer $\Rightarrow$ period $= d$. Period $= 1$ হলে **aperiodic**.

Example. $P_{01} = P_{10} = 1$ — এটাই চলতে থাকে:



n even $\rightarrow$ back করবে; n odd $\rightarrow$ back করবে না। $n \rightarrow \infty$ হলে odd/even নাই $\therefore$ limiting prob. exist করে না।

Limiting probabilities exist (and solve the equations above) for chains that are **aperiodic, irreducible** (and positive recurrent).

Lecture 15 Hardy–Weinberg Law & Gambler's Ruin 14/09/26

Hardy–Weinberg Law

Genotypes AA, aa, Aa with proportions p_0, q_0, r_0; $p_0 + q_0 + r_0 = 1$.

$$P(A) = P(A \mid AA)p_0 + P(A \mid aa)q_0 + P(A \mid Aa)r_0 = p_0 + \frac{r_0}{2}, \qquad P(a) = q_0 + \frac{r_0}{2}.$$

Offspring AA, aa, Aa with proportions p, q, r:

$$p = \left(p_0 + \frac{r_0}{2}\right)^2, \quad q = \left(q_0 + \frac{r_0}{2}\right)^2, \quad r = 2\left(p_0 + \frac{r_0}{2}\right)\left(q_0 + \frac{r_0}{2}\right)$$

মানুষের distribution change হয়ে গেছে ($p, q, r \neq p_0, q_0, r_0$ in general). Gene এর distribution:

$$P(A) = p + \frac{r}{2} = \left(p_0 + \frac{r_0}{2}\right)^2 + \left(p_0 + \frac{r_0}{2}\right)\left(q_0 + \frac{r_0}{2}\right) = \cdots = p_0 + \frac{r_0}{2}$$

$\Rightarrow$ **gene এর distribution same থেকে যায়** (as mutation নাই).

Claim: মানুষের distribution একবার change হয়ে (after one generation) unchanged থাকে।

Markov chain model (states AA, aa, Aa): একটা parent fixed রেখে আরেকটা parent all possible ways-এ combine (other parent is AA/aa/Aa w.p. $p/q/r$):

$$\mathbf{P} = \begin{array}{c|ccc} & AA & aa & Aa \\ \hline AA & p+\frac{r}{2} & 0 & q+\frac{r}{2} \\ aa & 0 & q+\frac{r}{2} & p+\frac{r}{2} \\ Aa & \frac{p}{2}+\frac{r}{4} & \frac{q}{2}+\frac{r}{4} & \frac{p}{2}+\frac{q}{2}+\frac{r}{2} \end{array}$$

$P_{AA\to AA}$**:** parent 1 AA (fixed), all possible parent 2:

- parent 2 AA (chance p): offspring AA w.p. 1;
- parent 2 aa (chance q): offspring AA w.p. 0;
- parent 2 Aa (chance r): offspring AA w.p. $\frac{1}{2}$.

$$P_{AA\to AA} = p\times 1 + q\times 0 + r\times\tfrac{1}{2} = p+\tfrac{r}{2}.$$

$P_{Aa\to aa}$**:** parent 1 Aa passes a w.p. $\frac{1}{2}$; parent 2 AA $\to 0$, aa $\to 1$, Aa $\to \frac{1}{2}$:

$$P_{Aa\to aa} = p\times 0 + q\times\tfrac{1}{2} + r\times\tfrac{1}{4} = \tfrac{q}{2}+\tfrac{r}{4}.$$

Note. Matrix fill-up করা আসতে পারে exam-এ।

According to the claim, $[\,p\ \ q\ \ r\,]$ is the limiting probability (π-matrix):

$[\,p\ q\ r\,]\,\mathbf{P} = [\,p\ q\ r\,]$ (we have to verify), e.g. first entry: $p\left(p+\frac{r}{2}\right)+r\left(\frac{p}{2}+\frac{r}{4}\right) = \left(p+\frac{r}{2}\right)^2 = p.$

The last equality holds because $p+\frac{r}{2}$ is the (unchanged) gene frequency.

Gambler's Ruin

Coin toss করতে করতে যতক্ষণ H পাব, তত টাকা (win \$1 w.p. p, lose \$1 w.p. $q = 1-p$). Depends on: initial কত টাকা in hand; coin এর bias কেমন।

Goal যদি N টাকা হয়, states $= N+1$ $(0, 1, \ldots, N)$.

$P_{00} = 1$, $P_{NN} = 1$ (একবার N goal পেলে terminate), $P_{i,i+1} = p = 1 - P_{i,i-1}$, $i = 1, \ldots, N-1$.

Let $P_i = P(\text{reaches } N \mid \text{starts with } i)$.

$$P_i = pP_{i+1} + qP_{i-1} \implies P_{i+1} - P_i = \frac{q}{p}(P_i - P_{i-1}), \qquad i = 1, \ldots, N-1.$$

With $P_0 = 0$, a **telescopic sum** gives

$$P_i - P_1 = P_1\left[\frac{q}{p} + \left(\frac{q}{p}\right)^2 + \cdots + \left(\frac{q}{p}\right)^{i-1}\right] \Rightarrow P_i = \begin{cases} \dfrac{1-(q/p)^i}{1-q/p}\,P_1, & \frac{q}{p}\neq 1 \\ i\,P_1, & \frac{q}{p}=1 \end{cases}$$

(P_1 we don't know). $i = N$ বসালে $P_N = 1$, which gives P_1:

$$P_1 = \begin{cases} \dfrac{1-q/p}{1-(q/p)^N}, & p\neq\frac{1}{2} \\ \dfrac{1}{N}, & p=\frac{1}{2} \end{cases} \qquad \therefore \qquad \boxed{P_i = \begin{cases} \dfrac{1-(q/p)^i}{1-(q/p)^N}, & p\neq\frac{1}{2} \\ \dfrac{i}{N}, & p=\frac{1}{2} \end{cases}}$$

Fair coin নিয়ে ঢুকলে pocket-এ যত টাকা, তার proportional.

$N\to\infty$**:** if $p > \frac{1}{2}$, $(q/p)^N \to 0$ so $P_i \to 1-(q/p)^i$ — but $i \gg$ হওয়া লাগবে (not realistic). If $p \le \frac{1}{2}$, $P_i \to 0 \Rightarrow$ **Ruin**.

Example. Patty starts with \$5, Max with \$10, $p = 0.6$ (for Patty). Mapping to the casino problem $\to N = 15, i = 5$:

$$P_5 = \frac{1-(2/3)^5}{1-(2/3)^{15}} = 0.87.$$

Lecture 16 Queueing Theory: Basics & Little's Law 19/09/26

Note. Mandatory set আসতে পারে (important).

Queueing theory: Customers $\to$ Server; waiting customers form a Queue. Server + Queue = **System** (capacity can be finite / infinite).

4 Key Quantities

(i)	L	avg. no. of customers in system
(ii)	L_Q	avg. no. of customers in queue
(iii)	W	avg. time a customer spends in system
(iv)	W_Q	avg. time a customer spends in queue

$$L > L_Q, \quad W > W_Q$$

W_Q small, W বড় $\Rightarrow$ queue-তে বেশিক্ষণ থাকা লাগে না, but service দিতে সময় বেশি লাগে। But পরের লোকগুলো queue-তে বেশি জমল না কেন? $\to$ মানুষের arrival rate কম।

Fundamental Cost Identity

λ_a = avg. rate of customers entering the system.

$$\text{avg. rate at which the system earns} = \lambda_a \times \text{avg. amount a customer pays}$$

Little's Law (by cost).
প্রতি second-এ pay করতে থাকলে (\$1 per unit time in system):

- প্রতি sec-এ টাকা দিলে, earning rate = L;
- মানুষ ঢুকার সময়েই (পুরোটা শুরুতে) W দিলে, system earns $\lambda_a W$.

$$L = \lambda_a W, \qquad L_Q = \lambda_a W_Q \text{ (শুধু queue consider করলে)},$$

$$\text{avg. no. in service} = \lambda_a \, \mathbb{E}[S] \quad (\mathbb{E}[S] : \text{ on avg কতক্ষণ service নেয়}).$$

Without cost (intuition, exam-এ explanation): system-এ W পরিমাণ সময় থাকে; পরের সবাই এসে queue grow করে। Service নিতে 10 sec, sec-এ 5 জন আসবে $\Rightarrow$ আমার জন্য $5 \times 10 = 50$ grow করবে।

Long-run Proportions

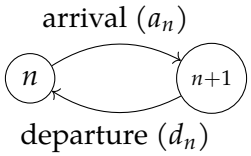
$X(t)$ = system-এ কতজন মানুষ আছে at time t.

$$P_n = \lim_{t\to\infty} P\{X(t) = n\} \quad \to \text{ total time এর কত অংশে } n \text{ জন ছিল}$$

$P_5 = 0.1 \Rightarrow$ overall time এর proportion (total time এর contiguous time-এ 5 জন ছিল — এমন না).

- a_n = যতগুলো arrival-এ customer এসে দেখবে n জন (proportion). 100টা arrivals এর মধ্যে 50 জন এসে দেখে n জন $\Rightarrow a_n = 0.5$.
- d_n = যতগুলো departure-এ (leaving customer) পিছনে n জন রেখে যায় (proportion).

$a_n = d_n$ always.



State = কতজন in system. Arrival edge 10 বার trigger হতে হলে departure edge-ও (প্রায়) 10 বার trigger হতে হবে: arrivals $\leq$ departures $+1$, $\therefore$ diff maximum 1 হতে পারে। But we're taking long-run proportions, so $a_n = d_n$.

PASTA (Poisson Arrivals See Time Averages). Arrivals গুলো Poisson process হলে, $P_n = a_n$. Poisson: memoryless $\Rightarrow$ unconditional (P_n) = conditional (a_n). (Slide: blue side.)

$\lambda = 5$: প্রত্যেক sec-এ 5 জন cust. আসে।

Lecture 17 The M/M/1 Queue

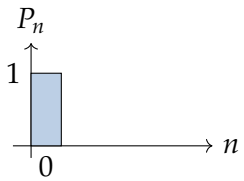
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Assumptions:

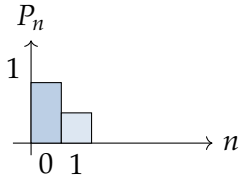
- Arrivals গুলো Poisson process (rate λ): events গুলো on avg λ rate-এ ঘটে; 2টা event এর মাঝখানে (avg) time $= \frac{1}{\lambda}$.
- Server প্রত্যেক unit time-এ (on avg) μ পরিমাণ customer-কে service দেয় (exponential service, rate μ).
- Single server, single queue.

$$\begin{array}{ll} \text{M} & \text{memoryless arrival} \\ \text{M} & \text{memoryless server} \\ 1 & \text{1 server} \end{array} \quad \Longrightarrow \text{ M/M/1}$$

System at state n = বর্তমানে কতজন customer. P_n: total time এর কত proportion-এ state n ছিলো। মাত্র created system: $P_0 = 1$ [আজীবনই 0th state-এ ছিলো], $P_{\neq 0} = 0$.



1 জন customer আসলে: যতক্ষণ 1-এ থাকবে, তত P_1 একটু বাড়বে, P_0 একটু কমবে। যে state থেকে বের হয়, তার $P_n \downarrow$; যে state-এ enter করে, তার $P_n \uparrow$.



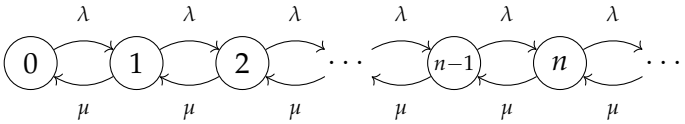
$$\frac{d}{dt}P_n(t) = \text{rate of entering } n - \text{rate of exiting } n$$

In the long run this becomes $0 \Rightarrow$ **"Balance equation"**:

> Rate at which the process *leaves* state n = rate at which it *enters* state n.

Balance eq. hold করেনি মানে system didn't converge to the long-run probabilities yet.





∞ পর্যন্ত বিস্তৃত: ∞ সংখ্যক state $\Rightarrow$ ∞ সংখ্যক balance equation.

Note. Link from the note: `tinyurl.com/irkqueue`

State 0: $\lambda = \mu$ (wrong). λ rate-এর কতটুকু অংশ 0-এর উপর effectively কাজ করে: λ এর P_0 অংশ (যতক্ষণ 0-তে) $= \mu P_1$ (যতক্ষণ 1-এ):

$$\lambda P_0 = \mu P_1.$$

> **Shortcut.** For each state: $\sum$(rate $\times$ prob. of the edge's start state) over outgoing edges = same over incoming edges.

State $n \geq 1$:

$$(\lambda+\mu)P_n = \lambda P_{n-1} + \mu P_{n+1} \;\Rightarrow\; \mu(P_{n+1}-P_n) = \lambda(P_n - P_{n-1}) \;\Rightarrow\; P_{n+1}-P_n = \frac{\lambda}{\mu}(P_n - P_{n-1})$$

Difference এর recurrence: let $D_n = P_{n+1} - P_n$, then $D_n = \frac{\lambda}{\mu}D_{n-1}$ and $D_0 = P_1 - P_0 = P_0\left(\frac{\lambda}{\mu} - 1\right)$. Telescoping,

$$P_n - P_0 = \sum_{k=0}^{n-1} D_k = P_0\left(\frac{\lambda}{\mu}-1\right)\sum_{k=0}^{n-1}\left(\frac{\lambda}{\mu}\right)^k \;\Rightarrow\; P_n = \left(\frac{\lambda}{\mu}\right)^n P_0.$$

Using $\sum_n P_n = 1$ (requires $\lambda/\mu < 1$):

$$P_n = \left(\frac{\lambda}{\mu}\right)^n\left(1-\frac{\lambda}{\mu}\right), \qquad n \geq 0$$

P_n & n এর relation: exponential (curve-টাও exponential).

> **Why $\frac{\lambda}{\mu} < 1$? (Exam).** $\lambda > \mu$: মানুষ fast আসে, service slow. Then $P_0 \cdot \infty = 1 \Rightarrow P_0 = 0, \therefore P_n = 0$: system $0 \to 1 \to \cdots \to 2 \to \cdots$ এভাবে অসীম states visit করবে, প্রত্যেক state-এ $P_n = 0$. $\frac{\lambda}{\mu} = 1$? converge করবে না, এত fast 0 হবে না।

> **Definition (Traffic intensity).** $\rho = \frac{\lambda}{\mu}$; stability condition $\rho < 1$. M/M/1 এর জন্য **utilization**-ও

বলা হয়। Server un-utilized: P_0 proportion; $1 - P_0 = \rho$: utilization.

$\mu > \lambda$ যত বড়, তত unutilized. Properly utilize: μ (সামান্য) $> \lambda$.

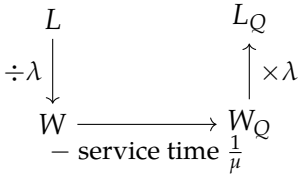
Performance Measures

$L = \mathbb{E}[n] = \sum nP_n$ — $n(\cdots)^n$ এর sum ঝামেলার (arithmetico-geometric), result:

$$L = \frac{\lambda}{\mu - \lambda} = \frac{\lambda/\mu}{1 - \lambda/\mu} = \frac{\rho}{1 - \rho}, \qquad \lim_{\rho \to 1^-} L = \infty$$

Trade-off. System-কে (fully) utilized রাখতে চাইলে on avg $L \to \infty$ — customer experience bad.

$$W = \frac{L}{\lambda} = \frac{1}{\mu - \lambda}, \qquad W_Q = W - \frac{1}{\mu} = \frac{\lambda}{\mu(\mu - \lambda)}, \qquad L_Q = \lambda W_Q = \frac{\lambda^2}{\mu(\mu - \lambda)}$$



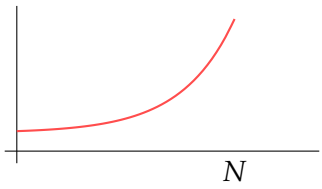
Exam recipe for a queueing model. P_n বের করব (balance equations) $\to L = \sum nP_n$ বের করব $\to$ Little's law for W, W_Q, L_Q.

M/M/1 Queue with Finite Capacity N

At most N in the system, i.e. $(N-1)$ জন at most line-এ থাকতে পারবে। States 0, 1 to $N-1$, $N \to 3$ types of balance eq. $\Rightarrow$ ...

$$P_n = \frac{(\lambda/\mu)^n(1 - \lambda/\mu)}{1 - (\lambda/\mu)^{N+1}}, \qquad n = 0, 1, \ldots, N$$

(no stability condition needed; if $\lambda > \mu$ the mass piles up near N).



λ rate-এ মানুষ ঢুকতে চায়, but N state-এ dropped. System এর point of view-তে arrival rate $\neq \lambda$; it is

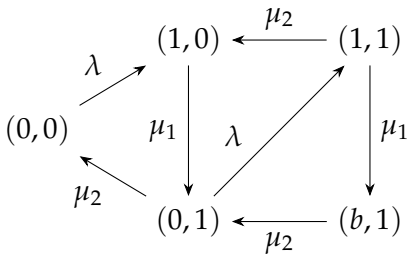
$$\lambda_a = \lambda(1 - P_N) \qquad (\text{“effective } \lambda\text{”}).$$

Lecture 18 Shoe Shine Shop, Tandem & Network Queues 26/09/26

Shoe Shine Shop

2 chairs: chair 1 $\to$ জুতা মেরামত (rate μ_1); chair 2 $\to$ shine করার (rate μ_2). Like an M/M/1 queue, but 2টা server (and no waiting room: a customer enters only if chair 1 is free).

States (i, j): i = chair 1 status, j = chair 2 status. **"Blockers":** $(b, 1)$ — chair 2 এর মানুষ service নিচ্ছে; chair 1 এর মানুষের service নেয়া done, but cannot go to 2.



Balance for $(0,1)$: $(\mu_2 + \lambda)P_{01} = \mu_1 P_{10} + \mu_2 P_{b1}$ — এভাবে all nodes. Also

$$P_{00} + P_{10} + P_{01} + P_{11} + P_{b1} = 1$$

(apply না করলে infinite solutions আসবে).

Actually enter করার effective rate:

$$\lambda_a = \lambda(P_{00} + P_{01}) \quad \text{(প্রত্যেক sec-এ on avg কতজন enter করছে)}$$

$$L = \mathbb{E}[n] = \sum nP_n = 0 \cdot P_{00} + 1 \cdot P_{01} + 1 \cdot P_{10} + 2P_{11} + 2P_{b1}, \qquad W = \frac{L}{\lambda_a}$$

(λ_a: enter করা মানুষের সাপেক্ষে নিব).

Proportion of customers who get blocked — Method 1:

$$\pi_b = \frac{\text{rate of chair-1 completions in state } (1,1)}{\lambda_a} = \frac{\mu_1 P_{11}}{\lambda(P_{00} + P_{01})}$$

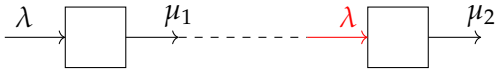
(numerator: প্রত্যেক sec-এ কয়টা blocker create; denominator: প্রত্যেক sec-এ কয়জন enter করে). Method 2: বই / কোথাও থেকে (see book).

Tandem Queue

Difference from the shoe shine shop: each server has infinite waiting room → waiting count infinite, তাই states infinite. State (n, m): n at server 1, m at server 2.

$$\underbrace{(\lambda + \mu_1 + \mu_2)P_{n,m}}_{\text{how to exit } P_{n,m}} = \underbrace{\mu_2 P_{n,m+1} + \mu_1 P_{n+1,m-1} + \lambda P_{n-1,m}}_{\text{how to enter } P_{n,m}}$$

(all nodes: slide). প্রত্যেকটাকে independent M/M/1 server ভাবলে: server 1 independent — trivial; server 2?



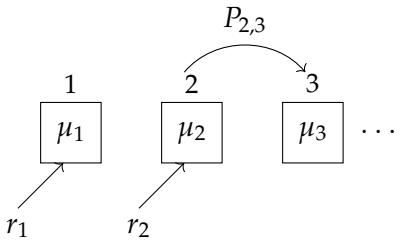
Server 2-কে independent হিসেবে model করতে পারব, যদি এই (input) rate $= \lambda$ হয়। E.g. hr-এ 2 জন আসে, server 1 hr-এ 100 জনকে service দেয় — server 2 এ effectively hr-এ 2 জন আসবে। λ is the bottleneck, not μ_1.

Product form.

$$P_{n,m} = \underbrace{\left(\frac{\lambda}{\mu_1}\right)^n \left(1 - \frac{\lambda}{\mu_1}\right)}_{\text{1ম server-এ } n \text{ জন}} \times \underbrace{\left(\frac{\lambda}{\mu_2}\right)^m \left(1 - \frac{\lambda}{\mu_2}\right)}_{\text{2য় server-এ } m \text{ জন}}$$

$(n, m) = (0,0), (0,1), (1,0), \ldots$ বসালে verify করা যাবে — "দেখাও যে, এরা independent." (Next blue slide — important.)

Open Network of Queues



r_j: server j-তে বাইরে থেকে arrival rate; P_{ij}: server i শেষ করে j-তে যাওয়ার probability. $\sum_j P_{ij} \neq 1$, unlike a Markov chain (customers can leave the system).

$$\lambda_j = \underbrace{r_j}_{\text{বাইরে থেকে}} + \underbrace{\sum_{i=1}^{k} \lambda_i P_{ij}}_{\text{অন্যান্য server থেকে}}$$

(λ_j: jth server-এ ঢুকতে পারে এমন total rate.) Equations solve করে λ গুলো find.

Caution. Total arrival rate হিসেবে $\sum_j \lambda_j$ বসালে $\rightarrow$ overcount (the same customer is counted at every server it visits); use $\sum_j r_j$ instead.

Note. Discuss, value বের করা — **mandatory**.